



Continuous countercurrent tangential chromatography (CCTC) for mixed mode post-capture operations in monoclonal antibody purification



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ABSTRACT

Continuous Countercurrent Tangential Chromatography (CCTC) has been shown to demonstrate significant advantages over column chromatography including higher productivity, lower operational pressure, disposable flow path, and lower resin use. Previous applications of CCTC have been limited to initial capture of monoclonal antibodies (mAb) from clarified cell culture harvest. In this present article, a CCTC system was designed and tested for a post-capture antibody purification step. Mixed mode cation exchange-hydrophobic interaction chromatography resins with two different particle sizes were used to reduce host cell protein (HCP), leached protein A, DNA, and aggregates from a mAb stream after a protein A operation. Product output from CCTC was obtained at a steady-state concentration in sharp contrast to the periodic output of product in multi-column systems. The results show up to 101 g of mAb/L of resin/hr productivity, which is 10× higher than in a batch column. A 5% yield increase (95% with CCTC vs. 90% in batch column) resulted from optimizing elution pH within a narrow operational window (pH 4–4.5). Contaminant removal was found to be similar to conventional column performance. Data obtained with the smaller particle size resin showed faster binding kinetics leading to reduced CCTC system volume and increased productivity. Buffer and water usage were modeled to show potential for utilization of in-line mixing and buffer tank volume reduction. The experimental results were used to perform a scale up exercise that predicts a compact CCTC flow path for 500 and 2000 L batches using commercially available membranes. These results demonstrate the potential of using CCTC for post-capture operations as an alternative to packed bed chromatography, and provide a framework for the design and development of an integrated continuous bioprocessing platform based on CCTC technology.

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1. Introduction

Monoclonal antibodies (mAbs) are the largest class of biopharmaceuticals and have driven much of the growth in the biopharmaceutical industry in the last twenty years [1]. Protein A column chromatography is usually used for initial capture of the mAb from complex cell culture fluid (CCF), with the subsequent purification accomplished using a series of post-capture chromatography steps, e.g., anion exchange, cation exchange, hydrophobic interaction, and mixed mode. However, conventional

column chromatography has multiple drawbacks including relatively high pressure, low productivity, and large capital costs. In addition, column chromatography is inherently a batch process, which necessitates innovation for the development of integrated continuous processes for lower cost and flexible biomanufacturing.

Continuous countercurrent tangential chromatography (CCTC) [2–5] has been used successfully for initial capture of mAb from CCF using Protein A affinity resins. In a CCTC process, the resin slurry flows through a cascade of static mixers and hollow fiber membranes while all of the chromatographic operations are performed simultaneously and at steady state. The static mixers are sized to provide appropriate residence times for all operations while the microporous hollow fiber membranes (0.65 µm pore size) retain

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the large resin particles (45–90 μm) allowing all dissolved species like proteins and salts to pass through into the permeate. The CCTC system operates at a low pressure (<70 kPa) in a continuous and steady-state mode. Other advantages of CCTC over batch column processes include uniform product quality, flexible manufacturing, higher productivity, and single-use operation.

The CCTC operation replaces the cyclic peak outputs that are common in batch and semi continuous multi-column processes with a straightforward steady-state product output that is easily monitored as a plateau by in-line UV. This can potentially lead to simpler process monitoring and improved product quality by providing the ability for feedback control of feed flow rate as well as precise control of product concentration during elution. In addition, CCTC employs a single-use flow path which largely eliminates the need for cleaning and cleaning validation. It is projected that more than 50% of all new commercial manufacturing facilities will be based on single-use technology, with a market size exceeding \$15 billion by 2020 [6].

Previous studies of CCTC for initial capture of mAb from CCF have shown productivities of up to 75 g of mAb/L of resin/hr, which is a 5–10 fold increase compared to batch columns [4,5]. A recent article by Dutta et al. [5] explored the optimization of CCTC systems for initial product capture, highlighting the trade-off between yield, productivity, and buffer usage and demonstrating the potential for further increases in system productivity by buffer recycling.

Whilst CCTC has been used for protein A capture, it has not been used for post-capture chromatography steps using mixed mode resins. The binding characteristics of these mixed mode resins are very different than those of protein A capture, both in terms of the binding kinetics/capacity and the sensitivity of the binding and washing operations to buffer conditions. The objective of this study is to provide the first demonstration of CCTC for post-capture operations of a mAb using mixed mode chromatography. The protein A capture step removed the majority of the HCP and DNA present in the cell culture fluid but the eluted product stream still contains significant amounts HCP, aggregates, and DNA in addition to leached protein A [7,8]. This work was focused on the removal of these contaminants using post-capture chromatography, which is commonly performed using cation exchange [9,10] or anion exchange [11,12] media.

Recent studies have shown potential advantages of multimodal or mixed-mode chromatography media (MMC) that exhibit multiple binding interactions [13,14]. For example, a combination of ion exchange and hydrophobic interactions can be exploited to increase the selectivity and purity for HCP removal in antibody purification [15–18]. In this study, experiments were performed with two N-benzyl-N-methylethanolamine containing resins with identical chemistries but different particle size (Capto™ adhere and Capto™ adhere ImpRes, both from GE), allowing us to also directly evaluate the effects of particle size on CCTC performance for the first time. A mAb with a known HCP removal profile using batch column with Capto adhere resin and with negligible levels of aggregate was selected for the current study, thus allowing a direct comparison of impurity removal between both batch and continuous modes of chromatography. The results clearly demonstrate the effectiveness of CCTC for post-capture operations, replacing traditional packed bed chromatography.

2. Materials and methods

2.1. Materials

A neutralized Protein A elution pool (post viral inactivation) was generated from a fed-batch CHO cell bioreactor and purified by Eli Lilly. The feedstock was stored at -80°C and then thawed and

sterile filtered (AseptiCap®, Advanced Microdevices, India) prior to use. Hollow fiber membranes from Spectrum Labs (98–127 cm^2 membrane area, 0.65 μm pore size, 0.75 mm ID, and 41 cm length, Rancho Dominguez, CA) were used for the CCTC flow-path. Post-capture was done using Capto™ adhere (referred to as Resin 1) and Capto™ adhere ImpRes (referred to as Resin 2) mixed mode chromatography resins with 70 and 36–44 μm particle diameter, respectively, both from GE Healthcare (Piscataway, NJ). Static mixers (L = 28.6 cm, ID = 1.0 cm) were obtained from Koflo Corporation (Cary, IL). All buffers were filtered using AseptiCap KS Polyether-sulfone Capsule Filters with 0.45 μm /0.2 μm pore size (Advanced Microdevices, India).

2.2. Equilibrium binding

5 mL of 14% resin slurry was combined with the mAb feedstock, and the resulting mixture was stirred continuously for 30 min. 1 mL samples of the fluid phase were obtained through 0.22 μm syringe filters (PES membrane, Advanced Microdevices, India), with the free antibody concentration evaluated using UV-A₂₈₀ using an Epoch spectrophotometer (BioTek, VT).

2.3. Batch binding kinetics

A batch system was used to study the binding kinetics. A known volume of resin slurry was rapidly mixed with a known volume of mAb feedstock. Approximately 1 mL samples of the fluid phase were collected at fixed time points through 0.22 μm syringe filters, with the mAb concentrations determined by UV-A₂₈₀. The mass of bound mAb was determined from the measured fluid phase concentrations using a simple mass balance.

2.4. Critical flux experiments

The permeate flux in the hollow fiber modules was maintained below the critical flux [19] to ensure stable operation of the CCTC system. The critical flux is defined as the permeate flux at which the Trans-Membrane Pressure (TMP) first becomes unstable, corresponding to the onset of fouling. The critical flux was evaluated as described in our previous publications [2–4]. Briefly, the resin slurry was circulated through a hollow fiber module in total recycle mode, with the retentate flow rate controlled using a peristaltic pump. Data were obtained by incrementally increasing the permeate and retentate flow rates every 20–30 min, with the ratio of the permeate to retentate flow rate (γ) maintained constant. Critical flux was evaluated by observing the first flow rate condition at which the pressure profile became unstable, as seen by a continuous increase in pressure during the constant flux filtration.

2.5. CCTC experiment

A schematic of the CCTC system used for this study is shown in Fig. 1. The process consists of five chromatographic steps: (1) binding, (2) wash, (3) elution, (4) strip, and (5) equilibration. The compositions of the buffers are shown in Table 1. The static mixers and hollow fiber modules were used as discussed earlier. The inlet and retentate (exit) flow rates were controlled using multi-head peristaltic pumps (Watson-Marlow, Wilmington, MA). The pressures in each step were monitored continuously using Pendotech Pressure sensors with data logging in LabVIEW. The system was first equilibrated with the system buffers. Resin was loaded to the system in the binding step after the pressures had stabilized. The outflow from the equilibration step was taken to drain until the exit resin slurry concentration from the equilibration step matched the stock resin concentration (14%); at this point, the resin tank was

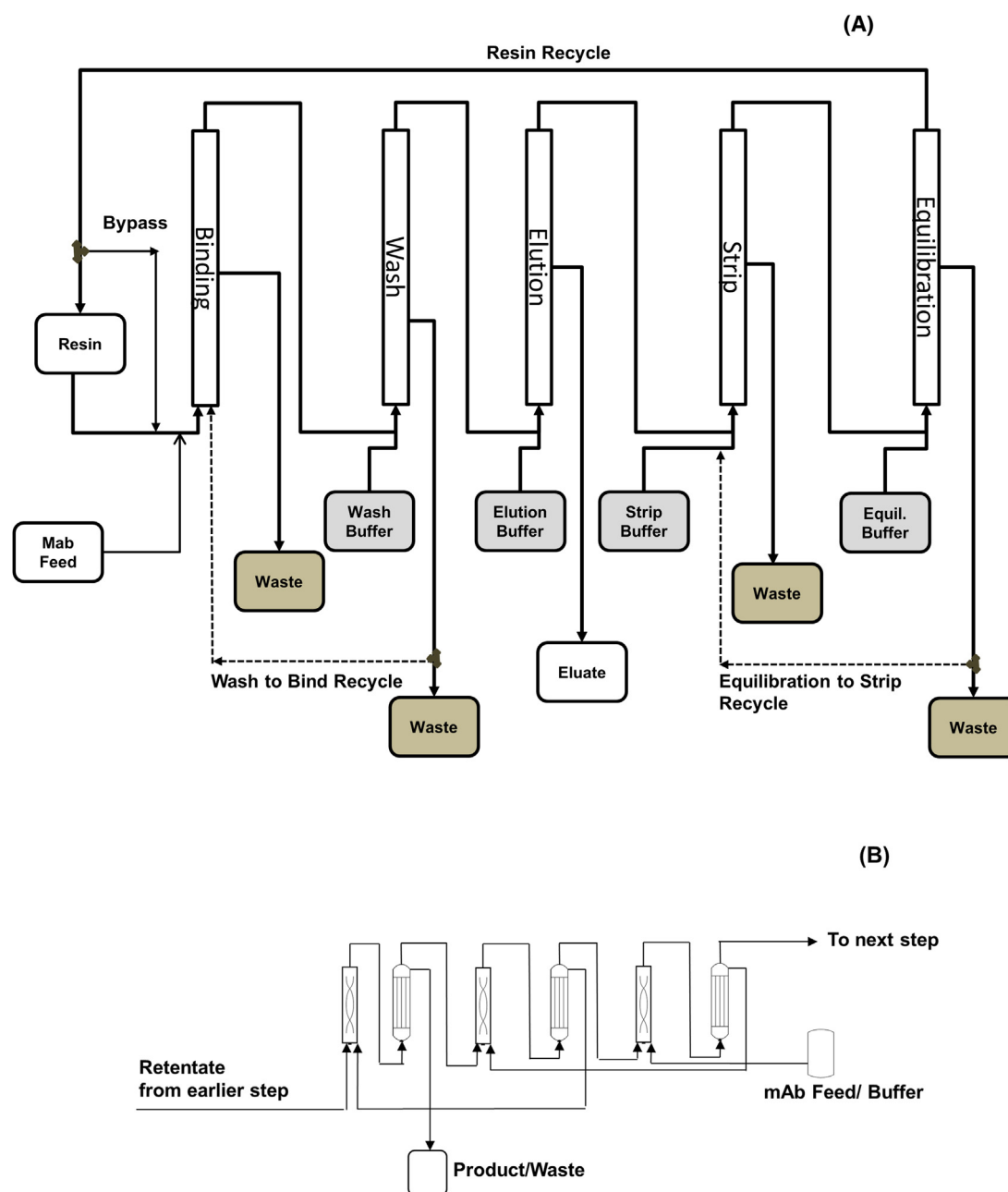


Fig. 1. (A) Schematic of the CCTC system operating in continuous recycle mode. The system consists of five steps: 1) binding, 2) wash, 3) elution, 4) strip, and 5) equilibration. (B) Schematic of a step showing 3 stages, each consisting of a static mixer and hollow fiber membrane module.

Table 1
Composition of different buffers used for this study.

Buffer	Composition	pH _{in}	pH _{out}
Bind	60 mM NaOAc	5.3	5.3
Wash	20 mM NaOAc + 100 mM NaCl	5.4	5.3
Elution	76 mM NaOAc + 50 mM NaCl	4.1	4.3
Strip	120 mM NaOAc + 500 mM NaCl	2.8	3.4
Equilibration	35 mM NaOAc	5.5	5.4

pH_{in} refers to the pH of the inlet buffer in the last stage of a particular step. pH_{out} is the pH of the solution coming out from the first stage of the same particular step except for equilibration step. For equilibration step, this is the pH of the retentate stream from the last stage.

bypassed (Fig. 1A) creating a closed system. The mAb load was then introduced to the system through the inlet to the binding step. All inlet streams (buffers and product) were filtered with 0.45/0.2 μ m capsules (AseptiCap KS, Advanced Microdevices, India) to reduce bioburden risk. The permeate samples from all steps were collected

to determine the HCP and protein A content as well as the mAb yield. Each step is made of multiple stages arranged in a countercurrent configuration to maximize performance [2]. A typical 3-stage step is shown in Fig. 1B.

2.6. Capto adhere column purification

A packed Capto adhere (multimodal anion exchanger, GE Healthcare) chromatography column was used for comparison with the CCTC system. The column was first pre-equilibrated with 20 mM sodium acetate, 1 M NaCl, pH 5.3 buffer for 2 column volumes (CV). The column was then flushed with 2 CV of 20 mM sodium acetate (NaOAc), 0.1 M NaCl, at pH 5.3. The product was applied to the column at various column loads, typically spanning a load range of 20–32 g of mAb/L of resin. After loading, the column was washed with 2 CV of 20 mM sodium acetate, 0.1 M NaCl, pH 5.3. Step elution was performed using 20 mM sodium acetate, 50 mM NaCl, pH 4.5. Pooling of the Capto adhere elution mainstream was done using in-line A_{280} absorbance cut points of 1.0 AU/cm on the front of the elution peak and 2.0 AU/cm on the back of the elution peak. The column was cleaned in place (CIP) after every cycle with 1 M NaOH using 2 CV and a total cleaning exposure time per cycle of 15–30 min. If the column was to be held >24 h prior to the next cycle, the column was stored in 0.01 N NaOH. All column steps were operated at a linear flow rate of 300 cm/hr.

2.7. Residual DNA

A quantitative polymerase chain reaction assay (qPCR) was used to determine the amount of residual DNA. This assay uses CHO-specific probes and primers to test for total CHO DNA in a sample after an extraction procedure to isolate the DNA. The quantitation limit is 60 pg/mL.

2.8. Size exclusion chromatography (SEC)

SEC HPLC was used to determine the monomer purity. A TSKgel G3000SWxl analytical SEC column (0.78 cm ID × 30 cm) was used with a mobile phase of 50 mM sodium phosphate, 300 mM NaCl. The mobile phase was applied to the column isocratically for 45 min at 0.5 mL/min. Quantitation was performed by UV absorbance using 214 nm and peak integration.

2.9. Residual host cell proteins

A sandwich ELISA developed by Eli Lilly was used to determine the amount of Chinese Hamster Ovary host cell proteins (CHO-HCP). The ELISA was standardized against a CHO-HCP laboratory reference standard made from null strain media supernatant. The quantitation limit for this method is 1 ng/mL.

2.10. Residual Protein A

A sandwich ELISA was used to determine the amount of residual Protein A. The ELISA was standardized against a recombinant Protein A reference. The quantitation limit for this method is 1 ng/mL.

3. Results and discussion

3.1. Design of the binding step

The static binding capacity (Q_m) was evaluated from batch uptake experiments based on the measured mAb concentration in free solution and appropriate mass balances:

$$Q_m = \frac{(C_0 - C_f) V_S}{\phi V_R} \quad (1)$$

$$V_S = V_R (1 - \phi) + 0.3 V_R \phi + V_M \quad (2)$$

where C_0 is the initial mAb concentration after mixing the mAb solution with the resin slurry (but before any binding to the resin),

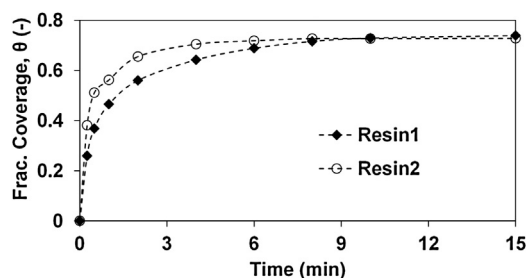


Fig. 2. Comparison of binding kinetics for the two resins. Resin 2 (Capto™ adhere Imp-Res) with its smaller particle size showed faster kinetics than Resin 1 (Capto™ adhere) because of lower diffusional mass transfer resistance.

C_f is the final concentration of mAb in solution (after binding equilibrium is reached), V_S is the total volume of the fluid phase in the final solution, V_R is the volume of the resin slurry, V_M is the volume of mAb feedstock, and ϕ is the resin concentration of the slurry (determined from the settled resin volume, v/v). The maximum binding capacities were 37.6 and 37.2 g/L for Resin 1 and Resin 2, respectively.

The ratio of the mAb feed flow rate to the resin slurry flow rate in the binding step was chosen as $\gamma_b = 0.6$ to achieve approximately 80% of the maximum binding capacity for both resins:

$$\gamma_b = \frac{LQ_m\phi}{C_m} \quad (3)$$

where L is the percent loading and C_m is the mAb concentration in the feed.

The binding kinetics for these two resins at this value of γ_b , which corresponds to the ratio of the volumes of the mAb feed to the resin slurry used in the batch binding kinetic experiments, are shown in Fig. 2. Resin 2 shows more rapid binding because of its smaller particle size, which significantly reduces the diffusive mass transfer limitations within the particles. For example, a fractional coverage of $\theta = 0.7$ was obtained after approximately 4 min with Resin 2 compared to more than 6.5 min for Resin 1. The more rapid binding kinetics allows the binding step to be designed with a significantly shorter residence time when using Resin 2. This was achieved by adjusting the static mixer volume used in the binding step, with $V = 370$ mL for Resin 1 and 240 mL for Resin 2.

3.2. CCTC design

The design equations for the CCTC system have been presented and discussed in detail elsewhere [4,5]. The value of γ (ratio of buffer flow rate to the concentrated resin slurry flow rate) was set at $\gamma = 3$ with 3 stages in the wash step (which should provide at least 98% removal of impurities). Details for the other steps are provided in Table 2. Typical data for the critical flux experiments at $\gamma_b = 0.6$ and $\gamma = 3$ are shown in Fig. 3. The critical fluxes for both resins were determined to be ≥ 650 L/m²/h at both the high and low values of γ . The membrane surface area for each hollow fiber module was chosen so that the permeate flux is less than 80% of the critical flux (giving an upper limit of 520 L/m²/h).

The number of stages, γ , and expected product yield (or washout factor) for the different steps in the CCTC process are given in Table 2 while Table 3 summarizes the key operating parameters for the two resins. The permeate flux used in the binding step was much lower than that used in the other steps due to significantly lower γ when compared with the other steps ($\gamma_b = 0.6$). Additional binding buffer was added after the static mixer (but prior to the membrane module) to increase the linear velocity and eliminate intermittent settling of resin particles. The ratio of binding buffer to resin flow rate was 1.7. The number of stages was based on the targeted frac-

Table 2

Design parameters for different steps.

Step	No. of stages	γ	J_{crit} (L/m ² /hr)	J_{Actual} (L/m ² /hr)	L_{Actual} (L/m ²)
Bind	1	0.6	≥ 650	256	889
Wash	3	3	≥ 650	404	4445
Elution	4	2.3	≥ 650	310	3408
Strip	2	3	≥ 650	312	3430
Equilibration	4	3	≥ 650	404	4445

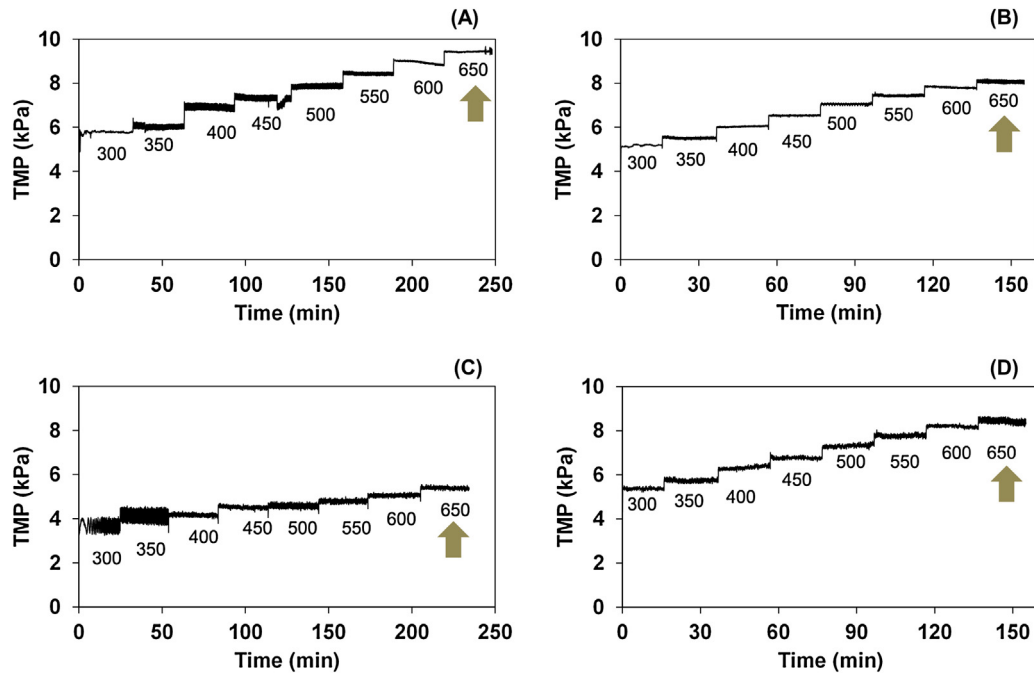


Fig. 3. Critical flux for Resin 1 at (A) $\gamma_b = 0.6$ and (B) $\gamma = 3$. Critical flux for Resin 2 at (C) $\gamma_b = 0.6$ and (D) $\gamma = 3$. The lower γ_b represents the conditions used in the binding step, while the higher γ is used in the other steps. The critical fluxes are marked with the arrows.

Table 3

Operating parameters for the CCTC runs.

Titer (g/L)	6.8
Q_{mAb} (ml/min)	13.2
Q_{slurry} (ml/min)	22
Resin (volume fraction, %)	14
$Q_{\text{buffer, total}}$ (ml/min)	286

tional impurity removal (wash, strip, and equilibration) or mAb yield (elution step) using the following equations [5]:

$$R(\text{or } Y) = \frac{\alpha(\alpha^N - 1)}{\alpha^{N+1} - 1} \quad (4)$$

$$\alpha = \gamma SK \quad (5)$$

$$K = \frac{1}{1 - \varphi_{\text{ave}}(1 - \omega)} \quad (6)$$

$$\varphi_{\text{ave}} = \frac{\varphi_{\text{in}}(\gamma + 2)}{2\gamma + 2} \quad (7)$$

where R is the fractional impurity removal, Y is the elution step yield, N is the number of stages in a given step, γ is the ratio of the buffer flow rate to the concentrated slurry flow rate, S is the average sieving coefficient for the species of interest ($S = 1$ for CHO cell proteins for the 0.65 μm pore size membranes used in the experiments), K is a correction factor that accounts for the presence of solids in the slurry, φ_{ave} is the average slurry volume fraction in the hollow fiber module (which is the mean of the inlet and outlet values), and ω is the liquid fraction contained in the packed (settled) resin bed ($\omega = 0.6$). For example, a wash step designed with

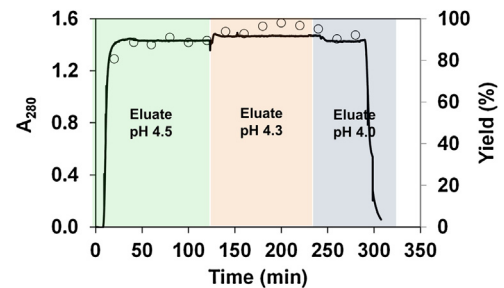


Fig. 4. A_{280} signal (black line) and yield (open circles) for the CCTC run using Resin 2.

three stages using $\gamma = 3$ and a 14% resin slurry inlet concentration should provide 97.7% impurity removal while a four stage elution step with $\gamma = 2.3$ should provide a mAb yield of 98.2%.

3.3. CCTC performance

Typical results for the mAb concentrations in the elution stream during a 5-h CCTC run are shown in Fig. 4. CCTC allows the user to seamlessly switch between different buffers while rapidly reaching steady state. In this run, the elution pH was optimized by sequentially utilizing three different elution buffers – pH 4.5 for the first 120 min, pH 4.3 for the next 110 min, and pH 4.0 for the final 75 min. The absorbance of the purified product clearly shifted because of these changes, corresponding to a mAb yield of around 90% at pH 4.5 and 4.0, with a yield of slightly more than 95% at the midpoint elu-

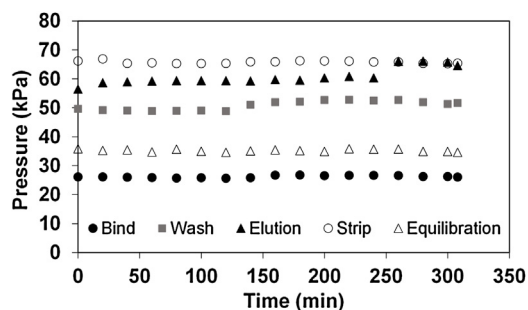


Fig. 5. Typical pressures in different steps during CCTC run using Resin 2.

tion pH of 4.3. The constant mAb concentration in the elution pool (at a constant elution pH) is in sharp contrast to the cyclical product concentration profile obtained in both simulated moving bed [20] and multi-column [21] chromatography systems. The results in Fig. 4 show that the CCTC system can easily be used to implement Design of Experiments (DOE) in a single CCTC run, and, in this case, to identify the optimal pH or the acceptable pH range for the elution step.

Table 4 compares the results obtained in CCTC runs using both the large and small particle size resins (Resin 1 and Resin 2, respectively) and a conventional packed column using the larger particle size resin. The steady state productivities of the CCTC system were 67 and 101 g of mAb/L of resin/h for Resin 1 and Resin 2, respectively. The higher CCTC productivity with the smaller size resin particle was a direct result of the smaller static mixer volume (i.e., smaller residence time) used in the binding step, which was enabled by the faster binding kinetics with the smaller size particles. The productivity with the smaller particle size resin is more than 10 times greater than that obtained using the column (9.7 g of mAb/L of resin/h).

Another way to analyze system productivity is to observe differences in cycle time, which for the CCTC operation is defined as resin residence time from binding through equilibration steps. The smaller particle size allows the CCTC system to cycle the resin considerably faster (~8 cycles per hour for Resin 2 vs. ~5 cycles per hour for Resin 1), compared with only ~0.3 cycles per hour for the corresponding batch column. It is important to note that a packed column could not be operated with the smaller particle size resin (Resin 2) at the same flow conditions due to the large expected pressure drop (the 2-fold reduction in particle size would be expected to give a 4-fold increase in pressure drop for operation with the same linear velocity and bed height), while such pressure limitations do not exist for CCTC since the particles are used in a flowing slurry. The small particle size Resin 2 could possibly be used in a conventional column but that would require operating at a lower velocity which would tend to reduce column productivity.

The mAb yield was higher for CCTC versus batch column (95% vs. 90%) possibly due to the pH optimization exercise as well as homogeneous mixing in the elution step. The HCP, leached protein A, and DNA (below detection limit), were all similar for CCTC and column operation, with approximately 1-log reduction in HCP. The monomer levels were slightly lower in the CCTC elution pool (by 0.2–0.5%), partially due to long term sample storage at low pH (0.2% drop measured in a controlled storage experiment at low pH) as well as the lower pH used in the elution step (pH 4.3 in CCTC vs. 4.5 in the column).

The stability of the CCTC system is examined in Fig. 5, which shows pressure profiles in the binding, wash, elution, strip, and equilibration steps throughout the 5-h run. The pressure profiles were nearly constant in all the steps, with deviations of less than 10% (except for a small increase in the pressure in the elution step after about 250 min which was caused by accidental pinching of

the permeate sample collection tube). All of the membranes used in Resin 2 were re-used for Resin 1 after a brief flow path sanitization (30 min with 0.5 N NaOH), demonstrating that it is possible to re-use the flow path to reduce membrane costs and/or to achieve longer process times. The maximum pressure observed in the CCTC system for these runs was less than 70 kPa (10 psi), which is much less than that observed in a typical column chromatography system (~300 kPa). Note that the estimated pressure required to run a column using the smaller resin particles would be more than 1200 kPa (170 psi) under the same flow condition as used for Resin 1 due to the large pressure losses associated with flow around such small particles.

Table 5 shows scale up modeling for processing 500 and 2000 L batches of the same feed when compared with the batch column operation. Because of the higher productivity of the CCTC system and the ability to use smaller resin particles, the resin usage is decreased by a factor of 8.3 (18.8 L for CCTC with Capto ImpRes vs. 156 L for the batch column with Capto Adhere for 2000 L processes). As shown in Table 5, the membrane sizing that would be required for these operations are commonly used in microfiltration operations and are available commercially.

The CCTC system allows seamless utilization of steady-state buffer recycling and in-line mixing with buffer concentrates because of the truly continuous nature of the process. These in-line mixing modules can be included in the single-use flow path, with each buffer line independently controlled by conductivity sensors with simple feedback control loops. Buffer usage in the tested set up was 3.4 L/g of purified mAb, while the batch column process used about 0.88 L of buffer/g of mAb. However, in a scaled up operation that utilizes in-line mixing, one could easily utilize 10× buffer concentrates for all steps, diluted with appropriate amounts of WFI which typically doesn't require additional storage capacity in the production suite. For the 500 L process, this would reduce the total buffer volume tankage for the CCTC system to less than 1100 L vs. 2900 L for a batch column without in-line mixing. It should be noted that in-line mixing systems do exist for batch columns; however, these are not widely used at manufacturing scale due to the challenging process control of these operations resulting from the non-steady state operating conditions.

Further reductions in water usage from 3.4 to 2.1 L/g (40%) are achievable with the CCTC process by recycling the wash effluent to the binding stage dilution stream and the equilibration effluent to the strip step (with additional spiking from a high salt/sodium acetate buffer) as shown by the dotted lines in Fig. 1. Similar recycling configurations were successfully tested in CCTC studies for capture chromatography. With these modifications a CCTC system processing a 500 L batch would still require 2.3× more WFI than a column. However the buffer tankage for this operation would be reduced even further to 690 L (more than a 4× reduction compared to a batch column) decreasing the necessary preparation time, vessel size, and needed floor space in the production suite. Further reductions in buffer use will be investigated in the near future through the use of innovative higher capacity and faster binding kinetics CCTC resins that will be uniquely suited for this application. These resins will have a significantly reduced particle size (20–25 µm) and softer/less crosslinked core structures, enabling much higher binding capacities and faster kinetics.

To test the effect of changes in flow rate on the system performance, a sensitivity analysis was carried out as shown in Fig. 6. Model simulations were performed for the elution step since this step has the greatest effect on the product yield. The pH of the retentate and permeate streams were calculated using CurTiPot software [22], while the change in elution step yield was determined using Eqs. (4)–(7). The results in Fig. 6A show that a ±10% change in elution buffer flow rate resulted in a negligible change in final eluate (retentate) pH (less than 0.02 pH units), with a somewhat larger

Table 4

Yield, Productivity and Product quality comparison between CCTC and Batch Column.

Performance	CCTC Capto Adhere (Resin 1)	CCTC Capto Adhere ImpRes (Resin 2)	Column*
Yield (%)	95	95	90
Steady state productivity (g/L/hr)	67	101	NA
Final elution pool Productivity (g/L/hr)**	55	81	9.7
HCP (ppm)	3	5	5
Leached protein A (ppm)***	0.6	0.7	0.2
Monomer in load (%)****	99.2	99.2	99.5
Monomer in elution pool (%)*****	98.7	99.0	99.6
Resin/process cycle per hour	5	8	0.3

Feed contained 48 ppm of HCP, 0.6 ppm leached protein A, 99.2–99.5% monomer.

*Column data were obtained using Resin 1 (Capto Adhere). Resin 2 (Capto Adhere ImpRes) was not used in column operations because of high pressure limitations.

** Process time used to calculate CCTC final elution pool productivity was based on the time period between starting CCF flow and stopping collection of elution sample.

*** Leached protein A values are within error limits of detection.

**** Monomer in load slightly lower for CCTC operation possibly due to prolonged storage and freeze/thaw.

***** Small loss of monomer in CCTC pool is partially due to long-term storage at pH = 4.3 before measurement (0.2–0.3%). Additional small loss may be due to pool pH that is slightly lower than column (pH = 4.3 vs pH = 4.5 for column).

Table 5

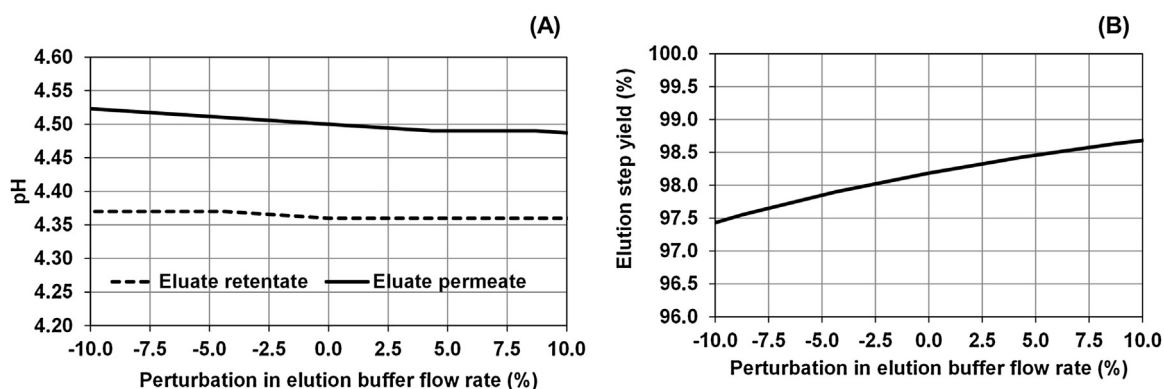
Scale up comparison between CCTC and Column operations for 9-h process time.

	CCTC (500 L)	CCTC (2000 L)	Column (500 L)	Column (2000 L)
Capto Adhere volume (L)	6.9	27.6	38.9	155.8
Capto Adhere ImpRes volume (L)	4.7	18.8	NA	NA
Average stage membrane area (m ²)	0.7–0.9	2.8–3.6	NA	NA
Total buffer volume (L)* (Buffer Concentrates in CCTC)	1083	4332	2926	11704
WFI volume (L)** (CCTC in-line mixing)	9750	39000	NA	NA
Total buffer volume with recycle (L)* (Buffer Concentrates in CCTC)	691	2764	NA	NA
WFI volume with recycle (L)** (CCTC in-line mixing)	6219	24876	NA	NA
Average Buffer Tank Volumes (L)***	150–250	600–1000	500–600	2000–2500

*Using 10× buffer concentrate for CCTC.

** WFI is assumed to not require additional tankage.

*** Average Buffer Tank Volumes are lower with CCTC because of steady-state in-line mixing.

**Fig. 6.** Sensitivity analysis showing the effect of elution buffer flow rate variation on (A) pH and (B) yield in elution step.

change in final permeate pH (0.02 units). The small reduction in pH with increasing buffer flow rate is due to the greater relative buffering provided by the low pH elution buffer. The same increase in the elution buffer flow rate will increase the elution step yield by slightly more than 0.6% due to the greater recovery of mAb eluted from the resin (Fig. 6B). Note that this ignores any change in elution kinetics with the very small change in elution pH. The mAb yield remains greater than 97.4% over this entire range of buffer flow rates.

In addition to HCP removal, post-capture unit operations are also expected to provide effective reduction of product related impurities such as aggregates. Since this study only demonstrated effective HCP reduction, our future studies will focus on assessing the effectiveness of the CCTC process described here in the reduction of product related impurities. Future studies are also being planned to examine viral clearance and CCTC system integration

with perfusion bioreactors with the vision of enabling steady-state end-to-end continuous bioprocessing.

4. Conclusions

This study provides the first demonstration of CCTC for post-capture operations, in this case using a mixed mode resin. The performance of the CCTC system was comparable to or better than a conventional packed column in terms of yield and impurity removal but with an 8–10 fold increase in productivity. The CCTC system was operated with a maximum pressure below 70 kPa (10 psi), enabling the use of a disposable plastic flow path suitable for implementation as part of a single-use flexible manufacturing system. Data obtained with the smaller particle size Resin 2 showed approximately 50% greater productivity due to the smaller residence time required for the binding step (but with no increase in

pressure drop for the flowing slurry and no loss of yield or impurity removal).

The CCTC system did use greater buffer volumes than a comparable batch column, in this case 3.4 L total buffer/g of mAb (compared to around 0.88 L/g for batch columns used for this type of post-capture step). However, because CCTC enables simple steady-state in-line buffer dilution, the buffer tanks needed for the operation would be significantly reduced when compared with column operation (by a factor of around 3–4). Modeling also shows that the CCTC buffer/water requirements can be reduced by 40% to 2.1 L/g by recycling buffers from the equilibration to strip and from the wash to binding steps. A comprehensive discussion of buffer optimization in CCTC is provided by Dutta et al. [5].

Scale up of the data shows that a CCTC system designed to process 500 and 2000 L batches would use membrane areas of 0.7–0.9 m² and 2.8–3.6 m², respectively, and volumes of the Capto Adhere ImpRes resin of 4.7 and 19 L, respectively. The membranes that would be required for this process are available commercially and would require no special modification for CCTC operations. It is anticipated that the CCTC process would provide significant economic savings versus the column operation, driven primarily by the large reduction in the required volume of the mixed mode resin and the corresponding increase in system productivity.

As shown by the elution pH optimization presented in this paper, the CCTC platform provides additional flexibility for performing in-line Design of Experiment (DoE) studies by changing buffer compositions and flow rates directly during a single experimental run. In this case, only pH was varied, but in principle, multiple variables could be changed to explore interactions between key variables. This would allow the user to optimize pH, flow rates, conductivity, loading and many other chromatographic and process variables through a suitably designed DoE.

The opportunity to perform post-capture operations using CCTC opens up the possibility of developing a fully continuous process for mAb production, moving directly from a perfusion bioreactor to a final drug substance.

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